Advanced SQL - Exercise 5
NULL Functions
Exercise 2.0

Q1. Show all employees with their salary. If salary is NULL, display 0.

```
SELECT employee_id,
       name,
       IsNULL(salary, 0) AS salary_with_defaul
FROM employees;
```

employees_id	Name	Salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2. Show employee name with their department. If department is NULL, show "Not assigned".

```
SELECT employee_id,
       name,
       IsNULL(department, 'Not Assigned')
FROM employees;
```

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Q3. find Orders with NULL customer_id using ISNULL.

```
SELECT order_id,
       customer_id
FROM Orders
WHERE customer_id IS NULL;
```

Order_id	Customer_id
103	NULL

Q4. Show all orders, if delivery_date is NULL show 'Pending'.

Order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

```
SELECT order_id,
       customer_id,
       ISNULL(CAST(delivery_date AS VARCHAR),
       AS delivery_status
FROM orders;
```

Q5. Show all students and their grades. Replace NULL with 0.

```
SELECT student_id,
       name,
       COALESCE (grade, 0) AS final_grade
```

FROM students;

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6. Counts students who haven't been graded.

```
SELECT COUNT(*) AS not_graded_count
FROM Students
WHERE grade IS NULL;
```

not_graded_count
1

Q.7. Show product name, price, and final price after discount (assume 0 if discount is NULL)

```
SELECT product_id,
       name,
       (price - COALESCE(discount, 0)) AS final_price
FROM products;
```

product_id	name	final_price
S01	Keyboard	25
S02	Mouse	10
S03	Monitor	100

Q8. Count how many customers have no email.

```
SELECT COUNT(*) AS missing_email_count
FROM customers
WHERE email IS NULL;
```

Missing_email_count
2

Q9. Show all customers with email. If NULL, display "no email".

```
SELECT customer_id,
       name,
       COALESCE(email, 'no email') AS email_display
FROM customers;
```

customer_id	name	email_display
1	Linda	no email
2	Joseph	joseph@mail.com
3	Nia	NO email

Q10. Show payment details with method replaced by "unknown" if NULL

```
SELECT payment_id,
       COALESCE(method, 'unknown') AS method_display,
       status
FROM payments;
```

payment_id	method_display	Status
301	Credit	NULL
302	Paypal	Success
303	unknown	failed

Q11. Show items and their quantity (0 if NULL).

```
SELECT item_id,
       item_name,
       COALESCE(quantity, 0) AS quantity_checked
FROM inventory;
```

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Q12. Show employee ID and the first available value among bonus or commission

```
SELECT emp_id,
       COALESCE(bonus, commission) AS first_available_reward
FROM employees_extra;
```

emp_id	first_available_reward
1	300
2	100
3	NULL

Q13. Count classes that don't have a room assigned

```
SELECT COUNT(*) AS no_room_count
FROM classes
WHERE room IS NULL;
```

no_room_count
2

Q14. Show attendance records with status, replace NULL with "Not marked"

```
SELECT student_id,
       date,
       COALESCE(status, 'Not marked') AS attendance_status
FROM attendance;
```

Student_id	date	attendance_status
1	2025-04-01	Not marked
2	2025-04-01	Present
3	2025-04-01	Absent

Q15. Show account ID, account_type (or 'unknown'), and balance (or 0)

```
SELECT account_id,
       COALESCE(account_type, 'unknown') AS type_display,
       COALESCE(balance, 0) AS balanced_checked
FROM Bank_Accounts;
```

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Q16. Show all projects with a start date. If start_date is null, display 'TBD'.

```
SELECT project_id,
       title,
       COALESCE(CAST(start_date AS CHAR), 'TBD')
       AS start_display
FROM projects;
```

project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Q17. Display reviews showing comment (or 'no comment') and rating (or 0)

```
SELECT review_id
       product_id,
       COALESCE(comment, 'no comment') AS
       comment_display,
       COALESCE(rating, 0) AS rating_display
FROM Reviews;
```

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

Q18. Show the supplier contact number. Use COALESCE (phone, alt_phone, 'No contact')

```
SELECT supplier_id,
       name,
       COALESCE(phone, alt_phone, 'No contact')
       AS contact_number
FROM supplier;
```

supplier_id	name	contact_number
1	Global Goods	123456789
2	Best Suppliers	987654321
3	Value Source	No contact

Q19. Show all users and their preferences. Replace all NULLs with defaults:

```
SELECT user_id,
       COALESCE (theme, 'light') AS theme_set,
       COALESCE (language, 'English') AS language_set,
       COALESCE (timezone, 'UTC') AS timezone_set
FROM user_settings;
```

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC +1
3	Light	English	UTC

Q20. Show maintenance log with:

```
SELECT record_id,
       machine_id,
       COALESCE(issue, 'unknown issue') AS issue_log,
       COALESCE(technician, 'not assigned') AS technician_name
FROM maintenance;
```

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not assigned
2	M102	Unknown issue	Not assigned
3	M103	Jammed	Alex